Research Statement

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My research program sits at the intersection of health economics, public policy, and applied microeconomics, united by a single question: how can rigorous quantitative evidence improve decisions that shape population health and economic well-being? With 25+ peer-reviewed publications cited nearly 1,000 times (h-index 15, i10-index 20), my work spans three connected domains — safety-net and food-system evaluation, public-health policy and behavioral economics, and global health economic evaluation — and speaks directly to agricultural economics, health economics, and health policy audiences alike. My methodological signature is the integration of modern causal inference with large-scale data analytics and decision-analytic modeling, always in service of questions policymakers actually face.

Research Agenda and Contributions

Safety-net programs, food systems, and economic shocks. A central pillar of my research evaluates how social safety-net programs affect health and economic outcomes, particularly during crises. My work on the Federal Pandemic Unemployment Compensation program (*Journal of Policy Analysis and Management*, 2024) showed that expiration of the $600/week supplemental benefit reduced food spending by 9.7% and food sufficiency by 6.0% — implying a marginal propensity to spend on food of 0.06 out of UI benefits, a key parameter for relief-program design. Using a control-function approach in an endogenous switching regression framework, I documented that smoothing effects were largest in high-unemployment states and among lower-income households, underscoring UI's countercyclical value. These findings informed national policy debates on pandemic relief.

My food-systems research uses Nielsen and IRI InfoScan retail scanner data comprising billions of transactions. In *Nutrients* (2021), I estimated structural demand systems (Exact Affine Stone Index) to quantify how SNAP participation shapes nutritional quality and spending patterns, documenting heterogeneity by store choice, income, and geography. Ongoing work estimates price and income elasticities across food groups to inform optimal SNAP benefit levels and food tax/subsidy design, and examines spillovers from energy assistance onto food security — an understudied channel with direct implications for integrated anti-poverty policy. This agenda connects naturally with agricultural and applied economics departments, where food policy, nutrition assistance, and consumer demand are core concerns.

Public health policy and behavioral economics. My co-first-authored study of COVID-19 vaccine lottery programs (*JAMA Network Open*, 2021; 79 citations) combined difference-in-differences with augmented synthetic control methods on 403,714 individual survey responses and state-level vaccination data. Lotteries were associated with a 2.1% increase in vaccine uptake overall; Ohio's $5 million program generated roughly 61,000 additional vaccinations at about $82 per induced vaccination — cost-effective relative to many public-health interventions. Effects were highly heterogeneous, proving ineffective in Arkansas, Kentucky, and West Virginia, highlighting how partisanship and baseline hesitancy condition incentive effectiveness. The study shaped national understanding of financial incentives in public health and was taken up by state health departments.

Chronic disease, health disparities, and risk prediction. In the *American Journal of Preventive Medicine* (2024), I analyzed 1.2 million adults from the BRFSS (2020–2022) on e-cigarette use and diabetes risk: sole e-cigarette users faced 7% higher odds of prediabetes (AOR 1.07),

dual users 24% higher (AOR 1.24), with effects concentrated among Hispanic users and users with obesity — findings directly relevant to FDA tobacco regulation. A companion decade-long analysis documented widening diabetes disparities by race/ethnicity, education, and geography. My co-first-authored machine-learning paper on diabetes prediction (*Int. J. Environ. Res. Public Health*, 2021; 301 citations) achieved 85%+ accuracy with random forests, XGBoost, and neural networks, influencing screening protocols internationally. Related NLP work (AgriBERT, IJCAI 2022; 81 citations) applies transformer models to agricultural and food systems, extended in a 2024 *IEEE Transactions on Big Data* paper investigating large language models for food and nutrition applications (55 citations).

Global health economic evaluation and decision modeling. As Health Economist and Data to Policy Lead at the CDC Foundation (contract to CDC), I lead economic evaluation under the Bloomberg Philanthropies Data for Health Initiative across 19 countries — designing cost-effectiveness, cost-utility, and budget-impact analyses and building Markov models, microsimulation, and decision trees for cancer screening, NCD prevention, maternal health, and road safety. I developed standardized CEA frameworks adopted by multiple LMIC health ministries, recently trained 37 participants from 19 countries in decision modeling (Cebu, Philippines, January 2025), and collaborate with WHO, the World Bank, and bilateral agencies on priority setting.

Works in Progress

Four active manuscripts extend the agenda above, and I will submit them as writing samples with faculty applications. **First,** "Monthly Cash, Food Spending, and Food Security: Evidence from the 2021 Child Tax Credit Expansion and Its Expiration" (with Sulakshan Neupane) exploits the American Rescue Plan Act's transformation of the Child Tax Credit into monthly payments (July–December 2021) and its abrupt expiration as a natural experiment, using a difference-in-differences design on Consumer Expenditure Survey and CPS Food Security Supplement data to estimate effects on food spending, food security, and the marginal propensity to spend, with heterogeneity by child age and income. **Second,** "SNAP Emergency Allotments and Food Security: Evidence from the 2020 Expansion and the 2023 Benefit Cliff" (with Sulakshan Neupane) exploits two sharp, large-scale SNAP changes — the March 2020 Emergency Allotments and the 2023 benefit cliff — to provide quasi-experimental evidence on how benefit generosity affects food security, with attention to rural households and benefit adequacy, directly informing Farm Bill debates. **Third,** "Rising Coronary Heart Disease Prevalence in the United States: Trends, Risk Factors, and Disparities, 2015–2022" (with Sulakshan Neupane, Uttam Dhakal, and Wojciech Florkowski; revise and resubmit) analyzes 3.5 million adults from the BRFSS using joinpoint regression for temporal trends and state-level geographic variation, extending my disparities research to cardiovascular disease. **Fourth,** "How Did the US Household Respond to the Economic Impact Payments of 2020? Evidence from the Consumer Expenditure Survey" (with Sulakshan Neupane) completes a stimulus-policy trilogy with my JPAM unemployment-insurance paper, estimating household spending responses to the 2020 direct payments.

Methodological Signature

What distinguishes my work is the fusion of econometric rigor with computational scale. I routinely analyze datasets with billions of observations, building purpose-built data pipelines (my small-area price-index work cut processing time by 50%). My toolkit spans causal inference (difference-in-

differences, regression discontinuity, instrumental variables, synthetic control, matching methods, endogenous switching regression), structural econometrics (demand systems, discrete choice), machine learning for prediction and heterogeneous-effect estimation, and decision-analytic modeling (Markov, microsimulation, QALY/DALY estimation). I work fluently in R, Python, Stata, SAS, SQL, and AWS — and I teach these tools, having served as instructor of record for "Analytical and Computational Tools for Applied Economics" at the University of Georgia.

Future Research Agenda

Over the next five years I will advance three priorities, each designed to yield high-impact publications while attracting extramural funding. **First,** health disparities and heterogeneous treatment effects: moving beyond average effects to examine how income, education, race/ethnicity, and geography moderate policy effectiveness, integrating machine learning for heterogeneous-effect estimation with quasi-experimental designs — aligned with NIH and AHRQ health-equity priorities. **Second,** causal inference with emerging data: high-frequency administrative data for sharper identification and text analysis of unstructured health data, pushing the causal-ML frontier in health economics. **Third,** global health economics and priority setting: standardized, data-sparse-adapted evaluation frameworks for LMICs (NCD, maternal-child health, infectious disease) with ministries of health, WHO, and the World Bank — alongside extended climate-adaptation research evaluating insurance, subsidies, and technology-adoption policies as health and food-security interventions (my Nepal work showed 21.6% revenue gains and 6.4% risk reduction from adaptation).

I will pursue funding from NIH, AHRQ, USDA NIFA, CDC, the Robert Wood Johnson Foundation, Bloomberg Philanthropies, the World Bank, and the Gates Foundation, and build a lab training doctoral students in the full pipeline from data engineering to policy translation.

Fit and Vision

My agenda is constructed to thrive in multiple departmental homes. For **agricultural and applied economics** departments: food-policy and nutrition-assistance research, structural demand modeling on scanner data, and climate-adaptation economics with field roots in Nepal. For **health economics** groups: economic evaluation, decision modeling, and disparities research with a global platform. For **health policy and public policy** schools: program evaluation that has informed Congressional debates and state health departments, plus demonstrated ability to translate technical analysis for decision-makers. Across all three I bring methodological innovation, large-scale data expertise, a strong publication trajectory, and commitment to mentoring the next generation.

The best econometric models are those that remember they describe human lives, not just data points. My research honors that principle — and I am eager to advance it as a faculty colleague.